

Depot

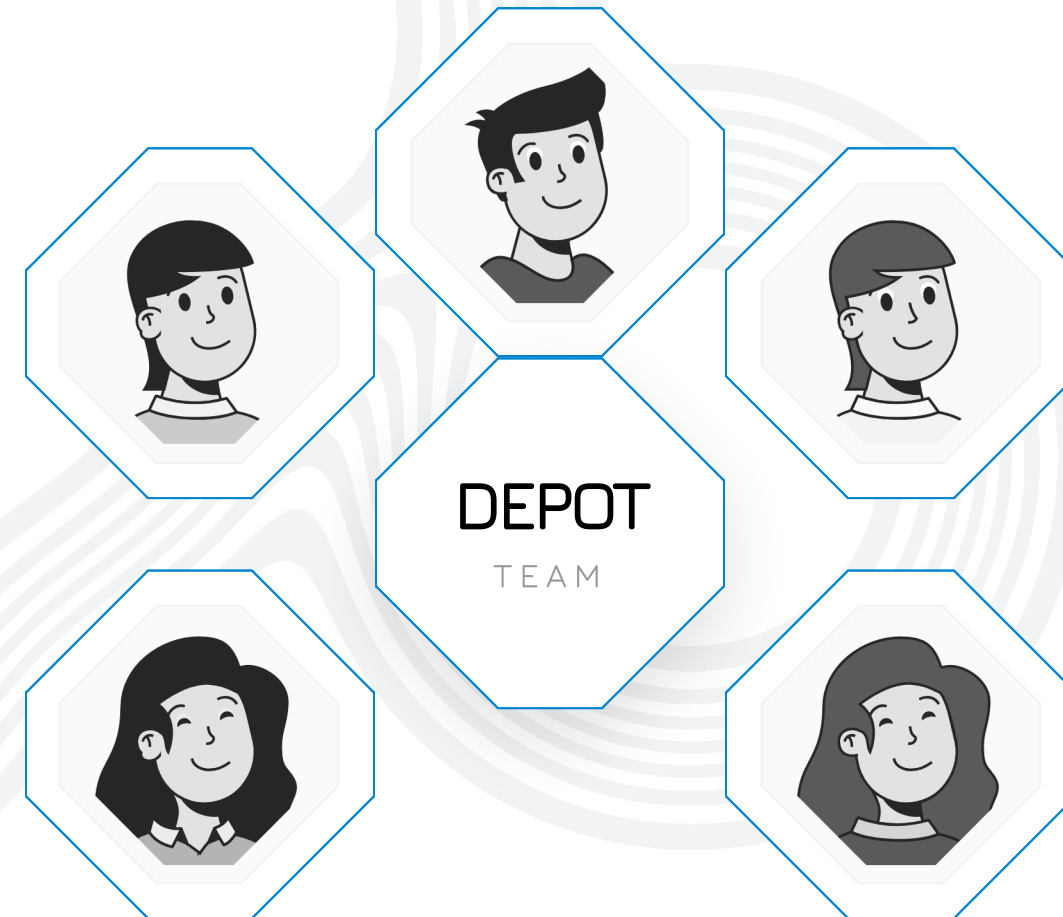
An infrastructure architect powered by AI,
from code to cloud.

TEAM

Ahsan Riaz
Muhammad Waleed
Muhammad Faizan Anwar

ADVISORS

Hira Anwar
Ayesha Hakim



Building apps is fast now. Shipping them is the bottleneck.

• Building is fast

~95%

of the code written by AI, at a quarter of YC's Winter 2025 start-ups

[TechCrunch, 2025](#) [1]

• Deploys break

~70%

of AI-written infrastructure failed to deploy on the first try, even from the best of six AI models

[Zhang et al., FSE 2026](#) [2]

Working code still leaves four decisions.

Decision 01



What it needs

Decision 02



Where it runs

Decision 03



What it costs

Decision 04



How to change it
safely

Depot answers all four, **and explains, records and approves every change.**

Each tool solves part of it.

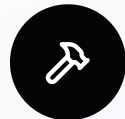
Someone still decides where it runs.

	Today Managed platforms Vercel, Railway, Render	Today Bring-your-own-cloud Porter, Qovery, Northflank, Ravion	Today Infrastructure-from-code Encore, Nitric, SST, Defang	Today AI for one provider or codebase Pulumi Neo, StackGen, Gemini Cloud Assist, azd init	Proposed Depot AWS, GCP, Azure; one gate for all
Starts from code as it is	✓ Yes	⊖ Partly	✗ No	⊖ Partly	✓ Yes
Chooses the cloud	✗ No	✗ No	✗ No	⊖ Partly	✓ Yes
Cuts cloud cost	✗ No	✗ No	✗ No	⊖ Partly	✓ Yes
Risk-assessed changes	⊖ Partly	⊖ Partly	✗ No	⊖ Partly	✓ Yes

Fifteen products, from their own documentation [3]. Partly = partly, or only some products; No = no, or not stated. Depot is planned.

Built for builders without an infrastructure engineer.

• User 01



AI-assisted builders

A working product, but little experience of running one.

• User 02



Solo full-stack developers

Outgrowing a managed platform: stay, or move to a cloud?

• User 03



Small teams and agencies

Two to ten people, with no DevOps staff.

Three objectives.

What a user can do with Depot.

01



Deploy any supported repository

Point Depot at a repo. It reads the app, picks the cloud, writes the infrastructure, prices it, provisions it.

02



Check health, metrics and cost

History, health, logs, spend and rollback in one place.

03



Remembers every decision

Why the architecture is what it is, what changed and when.

Research sits inside all three: comparing ten current AI models on architecture and IaC writing, and measured guardrails for the agent loop.

What Depot actually does.

Feature 01



Application Model

Every fact marked detected, inferred or user-supplied.

Feature 02



Architect

Designs the architecture and writes the Terraform from proven blueprints; self-verifies and auto-repairs.

Feature 03



Safety gate

Validate, saved plan, policy, cost and role check.

Feature 04



No stored keys

Bounded roles and federation; short-lived credentials.

Feature 05



Delivery and operations

Deploy on push, rollback, logs, cost, alerts, savings.

Feature 06



Agent and incident loop

Changes by conversation; alarms end in a fix a person approves.

One session, from working code to live.

Step 1



Connect

GitHub, GitLab or an
archive

Step 2



Answer

Up to eight questions

Step 3



Review

AI-written architecture
and Terraform, with
monthly cost

Step 4



Approve

Signed and role-
checked

Step 5



Run

Deploy on push, roll
back, watch cost

What Depot covers.

● In scope



GitHub, GitLab or
an archive



Apps, APIs, static
sites, workers



AWS, GCP, Azure



Short-lived
credentials

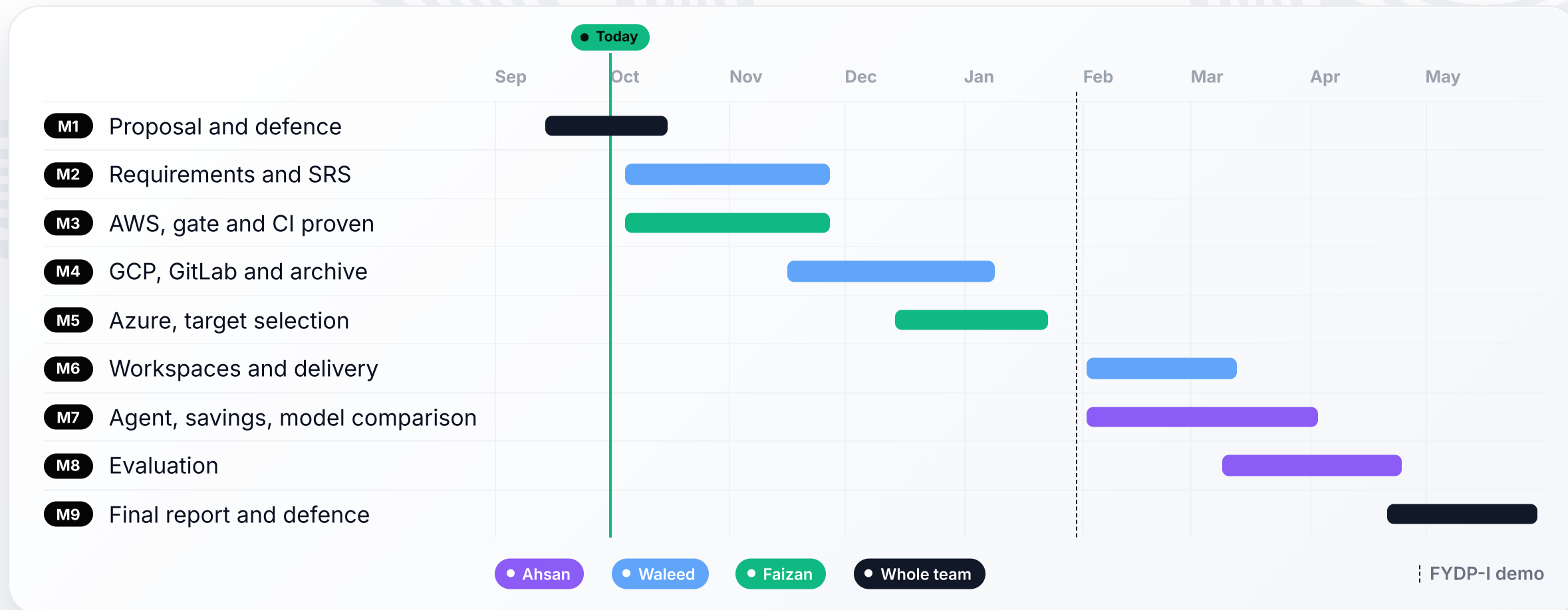


Approval before
every change



Extended scope only if M8 is met early: guided mode, more clouds (e.g. DigitalOcean, Oracle), voice, preview environments.

Three clouds first, then everything else.

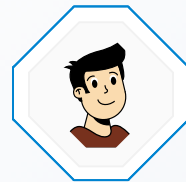


Team planning targets; dates will follow the official defence schedule.

The Architects

Three owners. No member reviews only their own work.

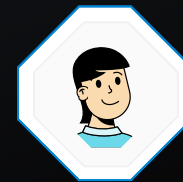
● 01 · 02 · 03



Ahsan Riaz

Application Model, agent mode, incident loop, research

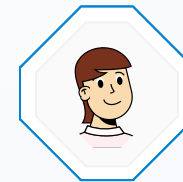
● 01 · 02



Muhammad Waleed

User-facing work, team workspaces, delivery and operations

● 01 · 03



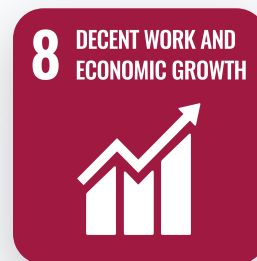
Muhammad Faizan Anwar

Architect, cloud adapters, safety gate, savings

Advisor **Hira Anwar** · Co-advisor **Ayesha Hakim**

Two goals we contribute to.

SDG 8.3



Small-enterprise growth

Small teams run products on cloud accounts they own, at a known monthly cost, without hiring.

SDG 9.5



Research and technological capability

An open benchmark, and views that explain why each architecture was chosen.

Sources.

- 1 [I. Mehta, "A quarter of startups in YC's current cohort have codebases that are almost entirely AI-generated," TechCrunch, Mar. 2025.](#)
- 2 [T. Zhang, S. Pan, Z. Zhang, Z. Xing and X. Sun, "Deployability-centric infrastructure-as-code generation: Fail, learn, refine, and succeed through LLM-empowered DevOps simulation," ACM FSE, 2026 \(arXiv:2506.05623\).](#)

- 3 Vendor documentation of the fifteen products compared.
- 4 [United Nations, "Goal 8" and "Goal 9," Sustainable Development Goals.](#)

Click a source to open it. Full references are in the proposal.

Thank you. Any questions?

• TEAM

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Ayesha Hakim

NUST SEECS · BS Computer Science